WORKSHEET · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 1: Algorithms: Precise Instructions

Module 1 | 45 minutes | Algorithms and sequence | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. algorithm ___ A. Find and fix bugs.
b. sequence ___ B. A mistake that makes a program behave wrongly.
c. bug ___ C. The order in which steps run.
d. debug ___ D. A precise set of steps to solve a problem.

2. Plan your code

Before you build it: Draw an “L”: start clean (erase all, go to x: -60 y: 40, point in direction 180 to face down), pen down, move 100 down the stem, turn left 90 to face right, then move 60 for the foot. Draw at least 2 lines. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Debug the square: this algorithm should draw a closed square, but one step is wrong. Find and fix it. Write or draw the blocks you will use.

4. How to debug

Put the debugging process in order Number them 1–5.

a. ___ Change one thing
b. ___ Find the first step that went wrong
c. ___ Compare what happened with what you expected
d. ___ Run the program
e. ___ Run it again to test

5. Define it

Which is the best definition of an algorithm? Circle one.

a. A. A precise sequence of steps that solves a problem
b. B. Any computer program
c. C. A type of computer

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can explain what an algorithm is.
☐ I can write a precise sequence.
☐ I can debug a sequence by testing one change at a time.